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IEC 61850 based energy management system using plug-in electric vehicles and distributed generators during emergencies

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ABSTRACT

During emergencies, alternative power system operation that utilizes local assets to energize local loads will change the face of the situation. While distributed generators (DGs) such as PV or wind turbines can be utilized for generation, plug-in electric vehicles (PEVs) can act as mobile generators. Different strategies for forming dynamic and networked microgrids utilizing the local generation during emergencies have been widely proposed in literature. However, the underlying communication required for optimal formation of dynamic microgrids and energy management during emergencies is not discussed. Addressing this gap, this paper develops a communication-based energy management system for emergency cases based on IEC 61850 standard. The IEC 61850 based communication modeling provides interoperability and serves for plug-and-play purposes. Furthermore, for establishing communication during emergencies the concept of Cognitive Radio (CR) is proposed in this scheme. Simulation results show the feasibility of modeling and acceptability of its performance.

1. Introduction

One immediate impact of global warming is more frequent occurrence of extreme weather events which impacts the normal power system operation. Several researchers have published their studies on disastrous impacts of extreme weather conditions on power system operations and losses in its aftermath [1–3]. These reports strongly highlight that power system's reliable operation in disastrous event and cyber/physical attacks on key electrical equipment are considered as main pillar for emergency recovery.

Recent events, such as Hurricane Irma, Mexico Earthquake, Fukushima Tsunami, demonstrated that current electrical systems and their operation principles are not fit for such events. There is a clear lack of detailed plan for continuous electricity supply under emergency conditions [4].

Vehicle-to-grid (V2G) technology is when an electric vehicle (EV) supplies power to support distribution network at peak times. With inverters that can operate in four quadrants, more auxiliary services can be provided such as frequency and voltage support, reactive power support as well as load factor adjustment. In this fashion, plug-in EVs (PEVs) can be utilized as energy sources in case of a natural disaster [5]. Assuming a moderate migration to EVs, V2G in a local neighborhood may have substantial impact on reliable and safe operation of power networks [6].

More in-line with the research presented in this manuscript, the distributed nature of PEVs and their availability near households, these PEVs could act as mobile generators to power critical loads through V2G support during emergencies. When a power network ceases its operation due to a hurricane, earthquake or terrorist attack, generators may be disconnected leaving many customers without electricity. Repairing lines that are ripped off or fixing substations that are rendered unusable may take long time. Even in a natural disaster at a very manageable scale, provision of relief and setting up a sustainable operation takes some time. In this case, an alternative power system operation that utilizes local assets to energize local loads will change the face of the situation. Based on the technologies above, it is possible to set up a microgrid or a single household system with available PEVs, by using V2G and vehicle-to-home (V2H), respectively [7].

Microgrids can be used as a solution for restoring power and enhancing resilience of power system during emergencies. In literature, different strategies for forming dynamic and networked microgrids during emergencies have been discussed [8–12]. In [8] authors propose a self-healing strategy, where the distribution system is optimally sectionalized into networked self-supplied microgrids during emergencies. For optimal formation of microgrids, a graph theory-based approach is proposed in [9]. Further, the optimal energy management in newly

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formed microgrids is also discussed in [9,10]. Similarly, authors in [11] propose a master slave approach for optimal formation of microgrids, where each island of microgrid consist of one master Distributed Generation (DG) to guarantee self-adequacy. However, in these studies, the underlying communication requirements for the optimal formation of dynamic microgrids is not discussed. In [12] dynamic microgrid formation optimization considering the information of remote DGs via multiagent wireless communication is proposed. The scheme utilizes transferring of simple ON/OFF messages for automatic control switches for formation of dynamic microgrids. However, the details of information exchanges and underlying wireless communication is not discussed. Also, the communication modeling and interoperability between all the equipment involved was not taken into account. Further, to run energy management optimization in dynamic microgrids during emergencies, more information related to available powers, dispatch etc., must be communicated among different components, which in turn requires interoperable and standardized communication. Addressing this research gap, this paper develops IEC 61850 standard [13] based interoperable and standardized communication model for power system energy management during emergency situations (PSMS-ES). The object-oriented modeling approach of IEC 61850 helps in organizing data, configuring objects and making them consistent and interoperable. There is a consensus among research and industry stakeholders that IEC 61850 will emerge as the communication standard of the future power systems [14].

During emergencies, for establishing communication it is expected that the available wireless networks and spectrum would be utilized. However, conventional wireless networks typically operate in a fixed spectrum range allocated by regulators. Hence, utilization of available spectrum and federating different available wireless networks is quite challenging. In this paper, this challenging task is realized by employing the Cognitive Radio (CR) concept which utilizes available spectrum and federate the communication of available coexistent networks during emergencies [15,16]. Further, IEC 61850 standard do not specify any particular protocol to be used at data link and physical layers. Thus, the concept of CR can be readily applied along with IEC 61850 communication without any issues of compatibility and feasibility. In this paper, implementation of IEC 61850 communication message exchanges for realizing energy management function during emergencies is carried out using IEC 61850 emulator tools demonstrating the viability of proposed communication model. Furthermore, performance evaluation of the proposed IEC 61850 communication based PSMS-ES using CR technology in terms of End to End (ETE) delay is presented. The main contributions of this paper are summarized as follows:

1. Develop IEC 61850 based communication model for Energy Management during emergency.
2. Implementation of IEC 61850 communication message exchanges using IEC 61850 emulator tools demonstrating the viability of proposed communication model.
3. Performance evaluation of CR based communication for the proposed IEC 61850 PSMS-ES scheme.

The rest of the paper is organized as follows: Section 2 gives a brief overview of power system management scheme for emergency cases (PSMS-ES), Section 3 shows the details of PSMS-ES communication and data links, Section 4 details the management algorithm utilized while Section 5 depicts its use with IEC 61850 based models and messages. Section 6 draws the conclusions.

2. Power system management scheme for emergency situations

In light of the above considerations, a Power System Management Scheme for Emergency Situations (PSMS-ES) has been developed earlier [17]. Developed scheme follows the flowchart given in Fig. 1. The overall idea is to shed all non-critical loads, detect healthy parts of the network along with all available generation units such as PV panels, wind turbines, storage units and PEVs. This approach tries to maximize the number of resources utilized for increased resiliency. However, if a building is completely isolated, the scheme falls back to stand-alone operation mode.

Fig. 2 shows different components that should be coordinated and utilized for PSMS-ES. The innovative approach in PSMS-ES proposes the use of PEVs and ferrying energy to places where it is most required, such as shelters or hospitals. The work in [17] documents the operation of PSMS-ES in detail; however, it does not develop the required communication modeling. This paper focuses on ensuring that different equipment from different vendors operate in a smooth way under emergency conditions. This interoperability is achieved by a standard approach. Therefore, IEC 61850 based approach is taken. Details of modeling, messaging and operation are given in the sections below.

3. PSMS-ES communication infrastructure

In a post-disaster scenario, PSMS-ES utilizes all PEVs and DGs, which are mobile and stationary, respectively, to feed the system with stable power output. In order to coordinate generation and demand-sides, it is necessary to employ intelligent energy management algorithms. In case of a disaster, most probably an interrupt occurs in electric supply and also ICT infrastructure, which is the key element for coordination and control of smart grid. Past experience in disaster management shows that phone services and email are not available while social media, such as Twitter or Facebook, is almost always up. Despite its drawbacks, social media can be utilized as a very strong tool to inform, organize and update the public during emergencies [18].

In addition to sharing information via social media, there is a communication infrastructure required to (a) achieve internet connectivity, (b) achieve connectivity between areas which may become islands after the disaster. Consider the case in Fig. 3, where a complete electrical network is torn apart and three post-disaster microgrids appear. To operate these with PSMS-ES as explained above, communication connectivity is vital. The helicopter and the drones shown represent data mules to keep the entire network as connected [19]. With their help, the emergency information can be passed on from one network to another and finally to crisis authorities. Through this approach, DGs, PEVs, and PSMS-ES are connected through a communication network. Considering the variety of above-mentioned devices, their duties and designs, such communication must be interoperable and flexible. Hence, to ensure interoperability and flexibility, the PSMS-ES communication is modeled as per IEC 61850.

Furthermore, it is important that such infrastructure is aware of its environment and can utilize different available spectrums and connect any existing wireless networks such as WiMAX, LTE, WiFi, GSM and radio networks. This challenging task during emergencies is realized by Cognitive Radio (CR) concept [15]. CRs are aware of their environment, internal states and can make intelligent decisions accordingly. CR networks (CRN) can behave according to their environments and support heterogeneity, reconfigurability, self-organization and interoperability [20]. All of these traits are equally valuable in safe-guarding connectivity after a disaster. Ideally, a disaster response network (DRN) needs to have Quality of Service (QoS), robustness, reliability, coverage, mobility, interoperability and self-organization.

The PSMS-ES functionally is embedded in the controllers of all dominant DGs in distribution system. If an emergency case occurs, the PSMS-ES will be activated in the dominant DG. Main responsibility of PSMS-ES is to coordinate and regulate the electrical energy flow in the microgrid. To achieve this, creating DRN is automatically undertaken. CR's sensing ability is utilized to detect active wireless networks and available spectrums. This detection is done locally to create a map of available communication devices and networks. If there is no active wireless network (incumbent devices) available/operating, DRN

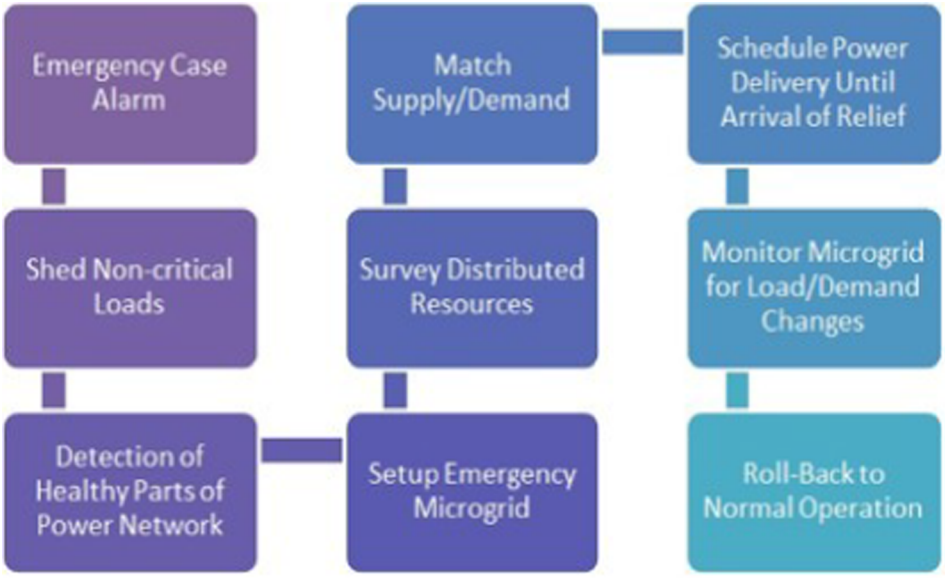


Fig. 1. Flow chart of PSMS-ES.

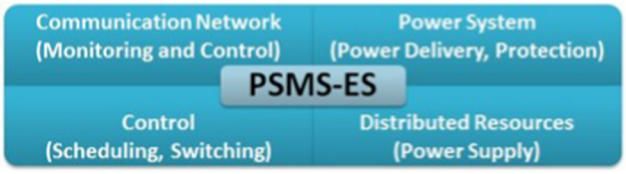


Fig. 2. Cyber-physical components of PSMS-ES.

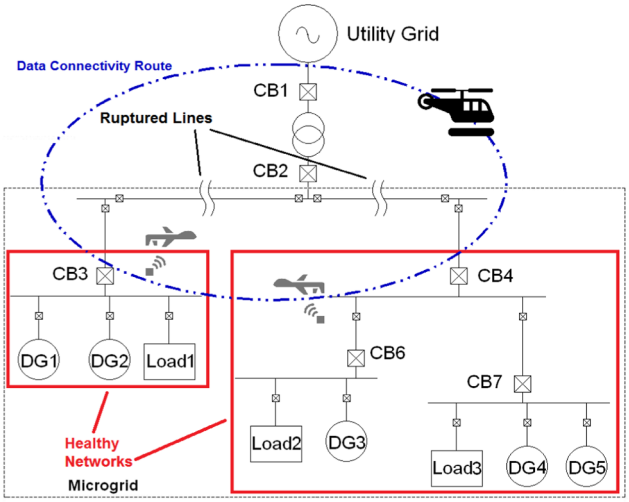


Fig. 3. Post Disaster configuration for achieving Connectivity in PSMS-ES.

operates in the spectrum freely (i.e. Interweave CR communication). If there are any incumbent devices operating in certain spectrum, the DRN chooses available free spectrum for its operation (i.e. underlay CR operation). Secondly, based on this information, CR creates an action plan which specifies the waveform that will be used and the communication links that will be established. In this fashion, DRN is formed with active links and available assets are brought together. Using the DRN, the PSMS-ES scans for available active power links and DGs in the vicinity by sending status request signals. The CBs/relays/switches on power lines respond to the PSMS-ES, which update the status of the power lines whether they are healthy of ruptured. The DGs responds to these signals from PSMS-ES with the information on their current status. When the PSMS identifies sufficient healthy lines and available DGs, healthy microgrid is formed and set operational. PSMS-ES uses DRN based on CRs to coordinate and regulate the electrical energy flow in the microgrid. Table 1 details the CR features that are required to create DRNs in healthy power networks. Although DRN is utilized for all sorts of communication and control, EV coordination is given as example in Table 1. Sensing feature makes sure that all EVs in the region are successfully detected. Planning is utilized to come up with a solid action plan so that PSMS-ES is seamlessly connected to PEVs. Finally, emergency case decisions made by PSMS-ES are relayed to EVs for mobilizing them to high-demand areas where they can be used as mobile generators.

Furthermore, the IEC 61850 standard do not specify any particular protocol to be used at data link and physical layers. The IEC 61850 messages can be used over any data link and physical layer protocols. Thus, the concept of CR can be readily applied along with IEC 61850 communication without any issues of compatibility and feasibility. IEC 61850 has emerged as one of the most promising solution for power utility automation since it is based on the interoperability approach. Different components of microgrids have been modeled as per IEC 61850 such as DGs [21,22], phasor measurement units [23], electric vehicles [24,25] and smart meters [26]. The information models of different components of microgrids developed in [21–26] are utilized in this paper to developed communication modeling of PSMS-ES.

Communication between DG units, PEVs, electrical loads and smart microgrid controller (PSMS-ES) can be achieved by exchanging IEC

Table 1
CR features and DRN needs.

Step No	CR Feature	DRN Need	Example
1	Sense/Observe	Self-Organization Detection of Spectrums Robustness	Detection of Available Assets
2	Plan/decision	Decision of Spectrum Selection of Services Connection to Nodes	Selection of a Spectrum that Ensures Robustness and determinism
3	Action /Coordination	Routing Info Sharing PSMS-ES Actions	Routing Messages sent by PSMS-ES to EVs (PEV & PHEV), for coordination

61850 MMS messages over TCP/IP communication and small cell technologies. The benefit of such approach is the ability to connect mobile phones and internet-based applications such as Smart Home Smart Charging, via one network. It has been reported in literature that such seaming is possible [20,27]. Scout Node concept has been developed in [27], where Scout Node can observe its environment, understand specifications and capabilities, list different channel models as well as policy and regulations, connect them with environmental realities such as geography, buildings etc. Finally, these can be utilized to undertake CR services.

In case of emergency, PSMS-ES system is operated independently by using available resources. There might be electric power outage and interruption of communication (especially in GSM service providers) simultaneously due to natural disasters or sabotages. PSMS-ES collects all the related information from the autonomous system from the microgrid environment. Redundant ICT infrastructure which consists of TCP/IP based networks and small cell technology allow all components of emergency microgrid to talk each other.

When DGs and electricity in the energy storage units is not sufficient to maintain the loads in the PSMS-ES area, it is necessary to employ additional PEVs which are located in the neighborhood. The PEVs equipped with the communication infrastructure can connect and coordinate with PSMS-ES. PSMS-ES gathers the geographic location and status of the SoC, SoH of each PEVs. Once the real time information is gathered from all DGs, energy storage units and PEVs, PSMS-ES coordinates the optimum power flow of the system. Available PEVs are called to the emergency microgrid to join a specific EV Cluster. PEVs serve as a DG under EV Cluster. This will help to balance fluctuated electricity generation from renewable energy sources. If the emergency case is over, this information will be transferred and PEVs will be relieved of their duties.

4. Energy management in PSMS-ES

This section presents required energy management functions to be carried out during the development of DRN once an emergency occurs. As soon as ES occurs, the PSMS tries to develop a schedule for the operation of the newly formed healthy microgrids for restoration of power to critical loads. The primary aim of PSMS, during ES, is to provide power restoration by keeping the operating limits and profiles of the available power sources. During ES, the controller of operating dominant generator in microgrid would act as the PSMS which is the authority to regulate the power flow in the microgrid by communicating with the DGs, loads and circuit breakers.

During the calamity, the power system network breaks down into 'm' microgrid islands. Each island contains various distributed generators (such as PV, diesel, PEV) and loads. The loads are further classified as critical and non-critical loads based upon the load priority. The PSMS during ES tries to supply power to loads in the order of priority based upon power availability from DGs. The main objective of the PSMS is to maximize the load supplied from the available power in the islanded microgrid over the time, t as (1)

$$MaxF = \sum_{\tau \epsilon t} w_m^l s_{m,t}^l \tag{1}$$

where, w^l is the weighting factor for loads based upon their priority and s^l is the status of the load. The weighting factor takes values as per (2) [28].

$$w^l = \begin{pmatrix} w^l = 1 \ \ for \ \ critical \ \ loads \\ w^l \leq 1 \ \ for \ \ non-critical \ \ loads \end{pmatrix} \tag{2}$$

The main objective of PSMS during ES is to satisfy the critical loads first. Hence the weighting factor is chosen as unity. If after supplying the critical loads, there is surplus power, then the non-critical loads are then supplied. The weighting factors of non-critical loads will be less than unity and non-critical loads will be supplied in the order of decreasing priority and respective weighting factors will be assigned resorting to non-critical load shedding.

The voltage and current limits in the tie lines must be within tolerance limit as given by (3)–(4)

$$V_i^{min} \leq |V_i(t)| \leq V_i^{max} \quad \forall \ t \in \tau \tag{3}$$

$$I_{ij}^{min} \leq I_{ij}(t) \leq I_{ij}^{max} \quad \forall \ t \in \tau \tag{4}$$

where V_i^{min}, V_i^{max}, I_{ij}^{min} and I_{ij}^{max} are the minimum and maximum line voltages and currents limits. The available power from healthy and operating DGs, batteries and EV cluster must be balanced to that of load demand. The net load demand is fulfilled based upon load priority as per the weighting factor introduced in (2). The net power balance is as per (5),

$$\sum_{i \epsilon \varsigma} P_{DER}^i(t) + \sum_{i \epsilon \beta} P_{bat}^i(t) + \sum_{i \epsilon N} P_{PEVdischarge}^i(t) = \sum P_l(t) \tag{5}$$

where $P_{DER}^i(t)$, $P_{bat}^i(t)$, $P_{PEVdischarge}^i(t)$ are the powers from the ith DER, energy storage and PEV joining the isolated microgrid island respectively and $P_l(t)$ is the power required by the loads.

$$\sum P_{DER} = \sum (P_{dispatchable \ \ DER} + P_{non-dispatchable \ \ DER}) \tag{6}$$

The limits of operation of DER must be within its maximum power supplying capacity, where P_{DER}^{max} is the maximum power supplied from a DER and is given by (7),

$$0 \leq P_{DER}(t) \leq P_{DER}^{max} \quad \forall \ t \in \tau \tag{7}$$

The battery charge level must maintain within its limits to maintain battery life where E_{bat}^{min} and E_{bat}^{max} are the minimum and maximum power limits of energy storage devices respectively and is given by (8),

$$E_{bat}^{min} \leq E_{bat}(t) \leq E_{bat}^{max} \quad \forall \ t \in \tau \tag{8}$$

The PEV capability constraints must also be adhered for reliable PEV operation as given by (9)–(10),

$$V_{PEV}^{min} \leq |V_{PEV}(t)| \leq V_{PEV}^{max} \quad \forall \ \tau \in t \tag{9}$$

$$I_{PEV}^{min} \leq I_{PEV}(t) \leq I_{PEV}^{max} \quad \forall \ \tau \in t \tag{10}$$

where V_{PEV}^{min}, V_{PEV}^{max} are minimum and maximum voltage limits of PEV respectively and I_{PEV}^{min}, I_{PEV}^{max} are minimum and maximum current limits of PEV respectively.

For supporting the power balance for supplying maximum power to loads in ES, the EVs act as mobile generators which could provide power to the microgrid islands as and when required. For this the PSMS assigns duties to EVs to join a respective EV cluster to supply power in a microgrid island. In this condition EVs acts in V2G mode.

An algorithm for energy management in PSMS-ES is developed in this paper for carrying out the load replenishment in emergency as shown in Fig. 4. The status of all DGs and breakers is fetched to the PSMS. Also, the status of loads based upon their criticality is made available at the PSMS through communication. Once this information is available, PSMS tries to formulate the energy management problem by maximizing the loads served with the available power from the DGs in the microgrid island. This energy management schedule is developed over a period of time for which it is valid. In the next time period, if emergency is not cleared, another energy management schedule is developed and conveyed to the respective DGs.

4.1. IEC 61850 Message Exchanges for energy management in PSMS-ES

The algorithm for implementing energy management during ES can be implemented by exchanging IEC 61850 based communication messages. The IEC 61850 information models available in literature for various DGs such as PV, Diesel etc., EV and various power system components [21–26] are utilized for developing information exchanges

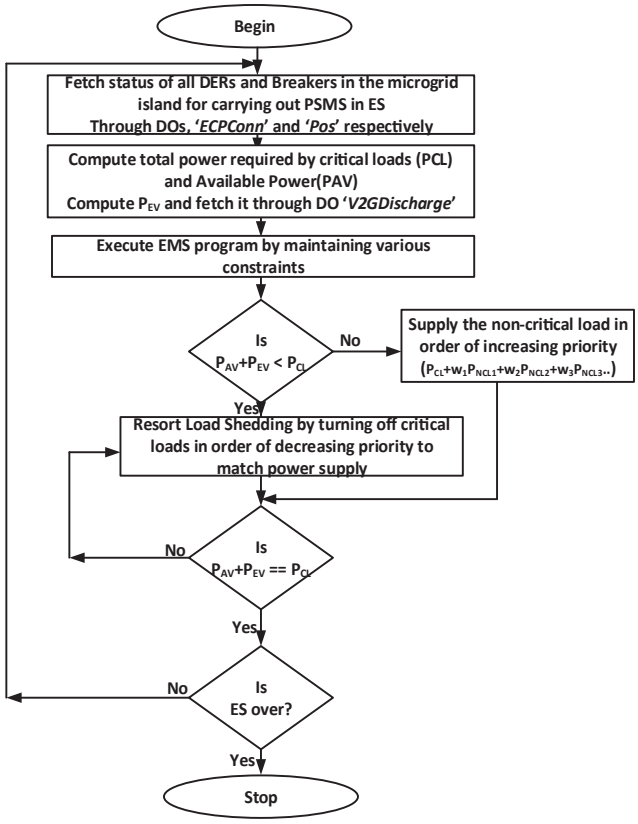


Fig. 4. Flowchart for energy management in PSMS-ES.

required.

For realizing the energy management algorithm in PSMS-ES, initially the status values of all DGs, circuit breakers, and loads are fetched at the PSMS. The status of DGs are made available through Logical Node (LN) DPST which contains a data object (DO) *'ECPConn'* which contains the value true (1) and false (0). Also, the available capacity of the DGs is fetched through the LN DRAT. The status of battery is fetched through LN ZBAT by various DOs such as '*MaxChaV*', '*AhrRtg*' for maximum charging voltage and ampere hour rating respectively. The status of circuit breakers is fetched to the PSMS through the LN XCBR and LN CSWI which contains the DO '*Pos*' for determining the normally open or normally close position of breaker. The LN CNLO [22] is used for fetching the status and characteristic of load.

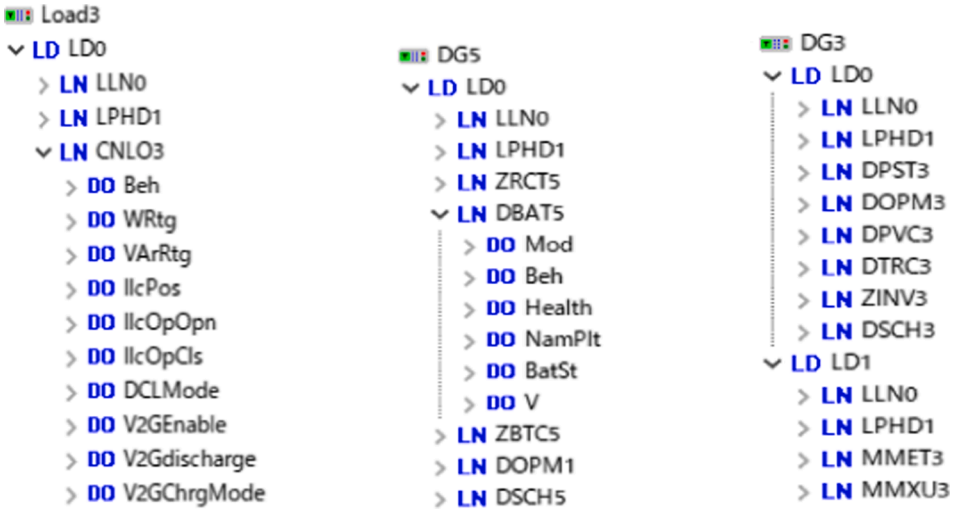
The EVs act as mobile generators for replenishing the power needs of microgrid whenever the power demand of critical loads exceeds the generation in that island. The EVs are then required to join the EV cluster and supply the power in V2G mode. When the EVs are connected, the DO '*V2GEnable*' in CNLO logical node of controllable loads is set to high. Other details regarding EV support at the controllable loads are conveyed through DO '*V2Gdischarge*' and '*V2GChrgMode*' for discharging power and charging operation of EV.

As soon as this information is made available at the PSMS, it calculates the available power supply and power demand in emergency situation. PSMS runs the optimization algorithms and calculates the energy service schedules and operating modes for different DGs. These schedules and operating modes are set in the DGs through logical nodes DSCH and DOPM. If available power is less than as required by critical loads, load shedding is resorted. Once the emergency is over, the energy management algorithm is stopped and normal power flow routine of microgrid is followed.

5. IEC 61850 implementation

The microgrid test system shown in Fig. 3 is considered for demonstrating the proposed communication based PSMS-ES. The microgrid test system consists of two dominant diesel generators DG-1 and DG-4. It is assumed that an emergency situation occurred and the feeders connecting to utility grid are damaged. This situation leads to formation of two healthy islanded microgrids as shown in Fig. 3 with some local generation. The healthy microgrid consists of three DGs (DG-3 PV plant, DG-4 diesel plant and DG-5 battery system), two loads (load-2 and load-3), where load-2 has EV integration capability, and three breakers (CB-4, CB-6 and CB-7) on three feeders.

For demonstrating the IEC 61850 information exchanges required for realizing the PSMS-ES, the ICD (IED capability description) files for all the components of microgrid have been developed using the InfoTech 61850 ICD editor tool [29]. The ICD file contains the semantics (i.e. IEC 61850 information model) of the particular IED. ICD files are required for emulating the communication capability of an IED. The CNLO logical node for controllable load proposed in [22] is used to develop information model in the form of ICD file. The instance of the ICD file developed for controllable load is shown in Fig. 5. Similarly, the information models for diesel plant, battery system, PV plant and breakers and their corresponding ICD files have been developed in this



(a) controllable load (b) Battery system (c) PV plant

Fig. 5. Information model different DGs from corresponding ICD files.

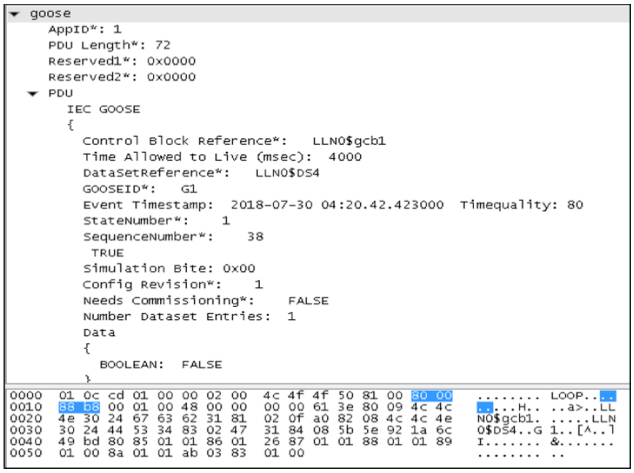


Fig. 6. GOOSE message capture published by PSMS to CB-4.

paper as shown in Fig. 5.

When ES arises, the PSMS-ES functionality is invoked in dominant generator DG-4. The PSMS-ES communicates with all the other DGs and loads for their availability through any available communication means as described in Section 3, such as CR, and thus forming a DRN. At first, the PSMS-ES sends a GOOSE command to trip of the CB-4 so that the healthy microgrid is islanded mode. The IEC 61850 GOOSE message exchange between the PSMS and CB-4 is implemented by emulating the PSMS and CB-4 over Avenue 2.0 and SCL Runner Infotech tools [29] respectively. The PSMS sets the status value (stVal) of DO 'pos' of the XCBR logical node in CB-4 to 'off' by publishing the GOOSE messages. Fig. 6 shows the GOOSE message published by the PSMS for opening the CB-4 captured by the Wireshark.

Now, the PSMS receives the information of all DGs, i.e. their status and ratings, through exchange of IEC 61850 MMS messages fetching data from appropriate logical nodes such as DRCT for DG-4, MMET and DPVC for DG-3, and ZBAT for DG-5. Further the information on criticality of each load is communicated to PSMS by the CNLO logical node. Since, load-2 has EV charging or discharging capabilities, in order to connect the EVs to load-2, the PSMS sets the DO '*V2GEnable*' of CNLO logical node to 'on' by sending a GOOSE message. Further, the load-2 communicates the amount of power the EVs can discharge to PSMS through IEC 61850 MMS message. This information is fetched from the DO '*V2Gdischarge*' in the CNLO logical node of controllable load.

After receiving the required information, the PSMS-ES performs the energy management optimization and generates different energy schedules for the DGs. These energy schedules for different DGs are communicated through IEC 61850 GOOSE messages to the DSCH and DOPM logical nodes which invokes the type of schedule or operating mode to be set in the DG. The DSCH and DOPM logical nodes are present in all the DGs. The schedule category and schedule type are set to 'emergency' and 'emergency islanding' by sending GOOSE messages carrying value '3′ and '9′ to the DOs '*SchdCat*' and '*SchdTyp*' in the DSCH logical node of all DGs. Different operating modes such as constant watts, constant voltage, constant vars, etc., obtained by PSMS from the energy management optimization function are set in the different DGs by publishing GOOSE messages. These GOOSE messages set the DOs corresponding to the appropriate operating modes in DOPM logical nodes to 'true', for example the DO '*OpModConW*' is set to 'true' for constant watts type operating mode, etc. Fig. 7 shows the GOOSE message published by PSMS containing status value for '*OpModConW*' as 'true'. This message is received by DG-4 logical node DOPM and operating mode constant watts is enabled. Similarly, the operating modes of the other DGs were set by sending appropriate GOOSE messages. Entire process is repeated after the expiry of the energy schedule period. Fig. 8 depicts the IEC 61850 communication model for microgrid with PSMS. All IEC 61850 messages exchanged for energy management functions in emergency situation are given in Table 2.

The performance evaluation of the proposed IEC 61850 communication based PSMS-ES using CR in terms of End to End (ETE) delay is carried out. Message exchanges listed in Table 2 for the microgrid test system are simulated over a CRN using the NetSim network simulator tool [30] as shown in Fig. 9. In the simulation, it is considered that all the CR nodes use the IEEE 802.22 standard operating in TV White Space (TVWS) range 54 to 698 MHz. The control IEDs (DG, load) are modeled with '*CR_CPE*' (consumer premises equipment) nodes available in NetSim library, which are secondary user CR nodes operating in TVWS 54–60 MHz range. The base station is modeled with '*Base_Station*' node operating in TVWS 54–66 MHz range. The other simulation parameters considered are listed in Table 3. Three scenarios are considered for performance evaluation. Scenario 1: there is no incumbent node present and the CR IEDs operate in the spectrum freely (Interweave CR communication). Scenario 2: An incumbent node operating in TVWS 54–60 MHz is present (overlay CR communication). Scenario 3: An incumbent node operating in TVWS 60–66 MHz is present (underlay CR communication). Incumbent node is added to the base station with ON duration of 10 sec.

The PSMS-ES functionality is invoked in DG-4 which exchanges messages with other DG-3, DG-5, load-2 and load-3. Custom application for each message exchange between PSMS-ES and each component is created in NetSim simulation.

The ETE delays for different message exchanges over the CRN is listed in Table 4. The time delays incurred at secondary user CR nodes, for spectrum sensing and identifying frequency holes, amount to significant portion of overall ETE delays in CRNs. Hence, the interweave CR communication (scenario 1) result in comparatively lower delays as there is no incumbent node present. Furthermore, it can be noted that ETE delay of GOOSE message tripping CB4 is less than the stipulated 3 ms for interweave and underlay CR communication. However, for overlay CR communication when incumbent node is operating in same frequency spectrum as secondary users (i.e. IEDs), the ETE delays for GOOSE messages is 5 ms which is not acceptable. However, the ETE delays of all the MMS messages exchanged for all the three scenarios are found to be well below the acceptable limits stipulated by the IEC 61850 standard.

6. Conclusions

There is growing awareness towards energy supply under emergency conditions. Whether it be increasing occurrence extreme weather events or threat of attacks on energy infrastructure, there are serious reasons to plan for contingency situations in electrical networks. The recent experiences, such as Hurricane Irma or Fukushima Tsunami, showed that there is a clear knowledge gap in this respect, and a dire need for a solution. A scheme has been developed by authors for continuous energy supply by making use of available assets and distribution lines in a local network. The energy is dispatched to last until the expected arrival of relief and makes use of PEVs which are becoming more prevalent. The important aspect of PEVs is their mobility and the ability to go to the demand, an unprecedented feature for supply/storage systems till date.

Emergency case microgrid is a perfect example of how Internet-of-things (IoT) can be a game changer and provide novel operation modes. It requires communication and control between very different devices. Achieving this in a standard and interoperable way is a daunting task. This paper develops IEC 61850 based operation to address this key knowledge gap in the literature. It ensures that different power system equipment, EVs as well as household items/loads can work together. This plug-and-play capability is achieved by combining IEC 61850 with smart grid and smart-city concepts. This standardized energy management system is very important for mitigating or minimizing the impacts of emergency cases on power networks, maximize the welfare of

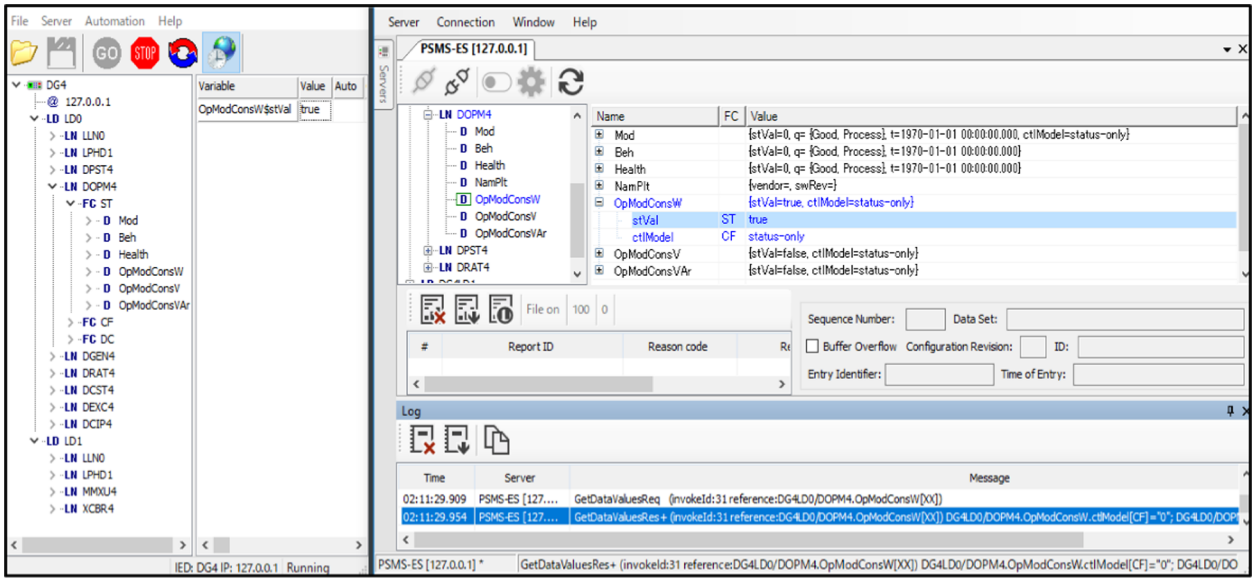


Fig. 7. Energy management schedule related IEC 61850 message exchanges between PSMS and DG-4.

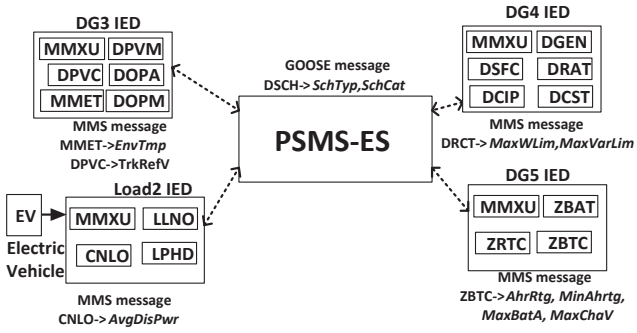


Fig. 8. Communication model of microgrid with PSMS-ES.

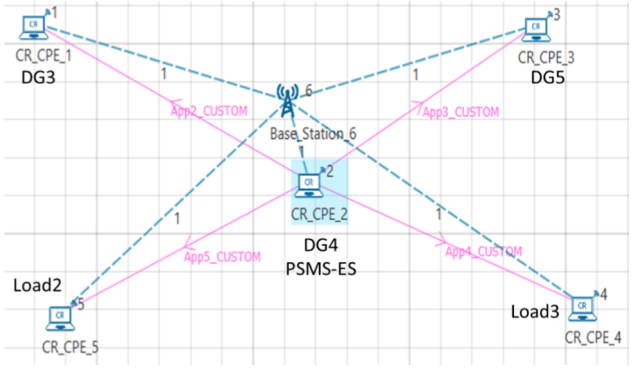


Fig. 9. CRN simulation of IEC 61850 communication based PSMS-ES.

affected communities and ensure seamless roll-back to normal operation once the situation turns to normal after relief arrives and/or emergency case ends. Furthermore, CR communication is proposed on top of IEC 61850 based communication model for PSMS-ES. Extensive simulation works were performed for information modeling and timing performance investigations for CRN in Infotech and Netsim software packages, respectively. Results show that developed models exchange the desired information successfully and show satisfactory timing performance. These show viability of the system in practical systems. As future work, auhtors would like to developed an optimized communication strategy with reduced IEC 61850 support and limited communication capabilities.

Table 2
IEC 61850 message exchanges for energy management in emergency situation.

Message Exchanges for Energy Management in ES		ACSI Service	Type of PDU/message
Source IED	Destination IED		
PSMS	CB4 → XCBR → *Pos,stVaL*	SendGOOSEMessage	GoosePDU
DG3 → MMET	PSMS	GetDataValues	Read-MMS Request
DG3 → DPVC			Read-MMS Response
DG5 → ZBTC → *AhrRtg, MinAhrtg, MaxBatA, MaxChaV*	PSMS	GetDataValues	Read-MMS Request
			Read-MMS Response
Controllable Load	PSMS	GetDataValues	Read-MMS Request
→CNLO			Read-MMS Response
PSMS	Load2 → CNLO → *V2GEnable*	SendGOOSEMessage	GoosePDU
Load2 → CNLO → *V2GDischarge*	PSMS	GetDataValues	Read-MMS Request
			Read-MMS Response
PSMS	All DGs → DSCH → *SchdTyp, SchdCat*	SetGoCBValues	Write-MMS Request
			Write-MMS Response
		SendGOOSEMessage	GoosePDU
PSMS	DG4 → DOPM → *OpModConW*	SetGoCBValues	Write-MMS Request
			Write-MMS Response
		SendGOOSEMessage	GoosePDU

Table 3
Simulation parameters.

Parameter	Value
Simulation Size	0.5 × 0.5 km
Protocol	IEEE 802.22
Multiple Access	OFDMA
FFT_Size	2048
Modulation	QPSK
DCD, UCD intervals	10 s
Uplink/downlink, ratio	4.5 Mbps, 1:1

Table 4
ETE delays for IEC 61850 communication based PSMS-ES over CRN.

Message exchanged between nodes	ETE in seconds		
	Scenario 1	Scenario 2	Scenario 3
PSMS → CB4	0.0025	0.005	0.003
DG3 ←→PSMS	0.36	0.74	0.36
DG5 ←→PSMS	0.36	0.74	0.36
Load2 ←→PSMS	0.358	0.714	0.358
Load3 ←→PSMS	0.358	0.714	0.358

CRediT authorship contribution statement

S.M. Suhail Hussain: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing - original draft. **Mohd. Asim Aftab:** Formal analysis, Investigation, Methodology, Data curation, Software, Validation, Writing - original draft. **Ikbal Ali:** Funding acquisition, Investigation, Project administration, Methodology, Resources, Writing - review & editing. **Taha Selim Ustun:** Funding acquisition, Investigation, Methodology, Project administration, Resources, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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